

13. Test Results and Observations

In API testing, **Test Results and Observations** is the final and most important phase of the testing lifecycle. It summarizes the overall execution of all APIs tested in this project using Fake Store API and provides insights into system quality, stability, and performance.

This section is used in **interviews, QA reports, and project documentation**.

13.1 What are Test Results?

Test results represent the **final outcome of executed API test cases**, showing whether each API request is:

- Passed ✓
- Failed ✗
- Skipped ⚠

It is derived from:

- Postman Collection Runner
 - Newman CLI execution
 - HTML reports
-

13.2 Test Execution Summary

After running the full API suite, we summarize:

◆ Total Test Cases Executed

Example:

- 30–50 API test cases (based on collection size)

◆ Passed Test Cases

- APIs responding correctly
- Valid status codes
- Correct response structure

◆ Failed Test Cases

- Incorrect response (if any bug exists)
- Missing fields
- Unexpected status codes

◆ Pass Percentage

Formula:

$$\text{Pass \%} = (\text{Passed Tests} / \text{Total Tests}) \times 100$$

13.3 Observations (Very Important Section)

Observations are **real-time findings during testing execution**.

◆ 1. Functional Observations

- All GET APIs (Products, Users, Cart) returned correct data
- POST APIs successfully created resources
- PUT and PATCH updated data correctly
- DELETE APIs removed records successfully

☞ Conclusion: Functional flow is stable

◆ 2. Data Validation Observations

- Response structure is consistent
- Required fields like `id`, `title`, `price` are present
- Data types are correctly maintained
- No null or malformed JSON observed

☞ Conclusion: Data integrity is good

◆ 3. Status Code Observations

- 200 OK → for successful GET requests
- 201 Created → for POST requests
- 404 Not Found → for invalid product IDs (expected behavior)
- 400 Bad Request → for invalid inputs

☞ Conclusion: Proper HTTP standards are followed

◆ 4. Performance Observations

- Average response time: under 2000 ms for most APIs
- Slight delay observed in Cart APIs in some cases
- No major performance bottleneck identified

☞ Conclusion: API performance is acceptable

◆ 5. Authentication Observations

- Login API generates token successfully
- Token works correctly for secured endpoints
- Invalid token scenarios handled properly

☞ Conclusion: Authentication system is working fine

◆ 6. Edge Case Observations

- Invalid product ID returns proper error
- Empty payload returns validation error
- Non-existing cart returns 404

☞ Conclusion: API handles negative scenarios properly

13.4 Defect Observations (If any)

If bugs are found, they are documented like:

Example:

- Product API sometimes returns missing category field
 - Cart update API not reflecting quantity immediately
 - Response time delay in heavy payload requests
-

13.5 Final Test Conclusion

After executing all APIs:

- Most APIs are stable and working as expected
 - CRUD operations are functioning correctly
 - Response validation and schema consistency are maintained
 - System behaves correctly for positive and negative scenarios
-

13.6 Project-Level Final Summary (Interview Answer)

In this project, I performed end-to-end API testing on Fake Store API using Postman and Newman CLI. I executed functional, regression, and integration test cases for Products, Cart, Users, Categories, and Authentication modules. I validated status codes, response bodies, JSON structure, and response times. After execution, I analyzed results using Collection Runner and Newman HTML reports, which confirmed that most APIs are stable, reliable, and production-ready.
